Worksheet 2.6 — Rational Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

The Big Ideas

Rational functions built from $\dfrac{a}{x-h}+k$ have asymptotes you can read straight from the equation.

- $y=\dfrac{1}{x}$ has asymptotes $x=0$ and $y=0$.
- $y=\dfrac{a}{x-h}+k$: vertical asymptote $x=h$, horizontal asymptote $y=k$.

Worked Examples

Example 1 — Reciprocal transformation

State the asymptotes of $y=\dfrac{4}{x-3}+2$.

Solution. Vertical: $x=3$. Horizontal: $y=2$.

Conclusion: $x=3$, $y=2$.

Example 2 — Reciprocal transformation with reflection

Sketch $y=\dfrac{-2}{x+1}-3$ as a transformation of $y=\dfrac{1}{x}$.

Solution. Reflected in x-axis and stretched by 2; shifted left 1 and down 3.

Conclusion: asymptotes $x=-1$, $y=-3$.

Example 3 — Domain of a general rational function

Find the domain and vertical asymptote of $y=\dfrac{2x-1}{x+5}$.

Solution. Denominator zero at $x=-5$.

Conclusion: domain all real x except -5; vertical asymptote $x=-5$.

Example 4 — Application

A club's average cost per member is $C(n)=\dfrac{600}{n}+15$ dollars, for n members. Interpret both asymptotes in context.

Solution. Vertical asymptote $n=0$: the model breaks down with zero members. Horizontal asymptote $y=15$: as membership grows, the average cost per member settles near \$15 (the per-person variable cost).

Conclusion: \$15 is the cost floor once fixed costs are spread thin.

Example 5 — Finding the equation from asymptotes and a point

A function $y = \dfrac{a}{x-h} + k$ has vertical asymptote $x = -4$, horizontal asymptote $y = -1$, and passes through $(-2, 3)$. Find its equation.

Solution. $y = \dfrac{a}{x+4} - 1$. Substitute $(-2, 3)$: $3 = \dfrac{a}{2} - 1 \Rightarrow a = 8$.

Conclusion: $y = \dfrac{8}{x+4} - 1$.

Example 6 — Simplifying a rational function with a common factor

Simplify $y = \dfrac{x^2 - 9}{x - 3}$ and state what happens at $x = 3$.

Solution. $\dfrac{(x-3)(x+3)}{x-3} = x + 3$ for $x \neq 3$.

Conclusion: the function simplifies to $y = x + 3$, but has a hole (not an asymptote) at $x = 3$, since that factor cancels.

Example 7 — Finding the x-intercept of a shifted reciprocal

Find the x-intercept of $y = \dfrac{5}{x-2} - 1$.

Solution. Set $y = 0$: $\dfrac{5}{x-2} = 1 \Rightarrow x - 2 = 5 \Rightarrow x = 7$.

Conclusion: x-intercept at $x = 7$.

Example 8 — Comparing two rational functions' asymptotes

State whether $y = \dfrac{3}{x-1} + 4$ and $y = \dfrac{-3}{x-1} + 4$ share the same asymptotes.

Solution. Both have the same $h = 1$ (vertical asymptote $x = 1$) and $k = 4$ (horizontal asymptote $y = 4$) — only the sign of a differs, affecting which quadrants the branches sit in.

Conclusion: yes, identical asymptotes; only the branch orientation differs.

Example 9 — Application — concentration approaching a limit

A drug's blood concentration is modelled by $C(t) = \dfrac{50t}{t+2}$ mg/L for $t \geq 0$. Rewrite in the form $\dfrac{a}{t-h} + k$ and interpret the horizontal asymptote.

Solution. $C(t) = \dfrac{50t}{t+2} = \dfrac{50(t+2) - 100}{t+2} = 50 - \dfrac{100}{t+2}$.

Conclusion: horizontal asymptote $y = 50$: the concentration approaches (but never reaches) 50 mg/L as time grows large.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

State the asymptotes of $y = \dfrac{1}{x+7}$.

Question 2

State the asymptotes of $y = \dfrac{5}{x} - 4$.

Question 3

State the asymptotes of $y = \dfrac{-3}{x-2} + 6$.

Question 4

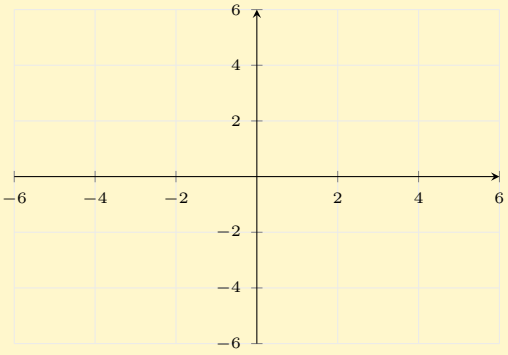
Find the domain of $y = \dfrac{x-1}{x+9}$.

Question 5

A function $y = \dfrac{a}{x-h} + k$ has asymptotes $x = 5$, $y = 1$, and passes through $(6, 4)$. Find a.

Question 6

Sketch $y = \dfrac{3}{x+2} - 1$, labelling both asymptotes.



Question 7

A gas's pressure is modelled by $P(v) = \dfrac{200}{v}$ for volume $v > 0$; explain what the vertical asymptote at $v = 0$ means physically.

Question 8

Simplify $y = \dfrac{x^2 - 16}{x - 4}$ and state what happens at $x = 4$.

Question 9

Find the x-intercept of $y = \dfrac{6}{x+1} - 2$.

Question 10

A function $y = \dfrac{a}{x-h} + k$ has vertical asymptote $x = 3$, horizontal asymptote $y = 2$, and passes through $(5, 5)$. Find its equation.

Question 11 — Challenge

A function $y = \dfrac{a}{x-h} + k$ has asymptotes $x = -3$ and $y = 2$, and passes through $(-1, 5)$. Find a, then find the x-intercept.

Question 12 — Challenge

Rewrite $C(t) = \dfrac{80t}{t+4}$ in the form $k - \dfrac{a}{t+4}$ and state the horizontal asymptote.

Question 13 — Challenge

A function $y = \dfrac{a}{x-h} + k$ passes through $(0, 5)$ and $(4, 9)$ with vertical asymptote $x = 2$; find a, k.

Question Recap & Answer Key

Question Recap

1. State the asymptotes of $y = \dfrac{1}{x+7}$.
2. State the asymptotes of $y = \dfrac{5}{x} - 4$.
3. State the asymptotes of $y = \dfrac{-3}{x-2} + 6$.
4. Find the domain of $y = \dfrac{x-1}{x+9}$.
5. A function $y = \dfrac{a}{x-h} + k$ has asymptotes $x = 5$, $y = 1$, and passes through $(6, 4)$. Find a.
6. Sketch $y = \dfrac{3}{x+2} - 1$, labelling both asymptotes.
7. A gas's pressure is modelled by $P(v) = \dfrac{200}{v}$ for volume $v > 0$; explain what the vertical asymptote at $v = 0$ means physically.
8. Simplify $y = \dfrac{x^2 - 16}{x - 4}$ and state what happens at $x = 4$.
9. Find the x-intercept of $y = \dfrac{6}{x+1} - 2$.
10. A function $y = \dfrac{a}{x-h} + k$ has vertical asymptote $x = 3$, horizontal asymptote $y = 2$, and passes through $(5, 5)$. Find its equation.
11. A function $y = \dfrac{a}{x-h} + k$ has asymptotes $x = -3$ and $y = 2$, and passes through $(-1, 5)$. Find a, then find the x-intercept.
12. Rewrite $C(t) = \dfrac{80t}{t+4}$ in the form $k - \dfrac{a}{t+4}$ and state the horizontal asymptote.
13. A function $y = \dfrac{a}{x-h} + k$ passes through $(0, 5)$ and $(4, 9)$ with vertical asymptote $x = 2$; find a, k.

✓ Answer Key

1. $x = -7$, $y = 0$
2. $x = 0$, $y = -4$
3. $x = 2$, $y = 6$
4. all real x except -9
5. $4 = \dfrac{a}{1} + 1 \Rightarrow a = 3$
6. asymptotes $x = -2$, $y = -1$
7. As volume approaches 0, pressure would grow without bound — physically, a gas cannot be compressed to zero volume
8. $= x + 4$ for $x \neq 4$; hole (not asymptote) at $x = 4$
9. $\dfrac{6}{x+1} = 2 \Rightarrow x + 1 = 3 \Rightarrow x = 2$
10. $5 = \dfrac{a}{2} + 2 \Rightarrow a = 6$: $y = \dfrac{6}{x-3} + 2$
11. $5 = \dfrac{a}{2} + 2 \Rightarrow a = 6$; intercept: $0 = \dfrac{6}{x+3} + 2 \Rightarrow x + 3 = -3 \Rightarrow x = -6$
12. $C(t) = 80 - \dfrac{320}{t+4}$; horizontal asymptote $y = 80$
13. $5 = \dfrac{a}{-2} + k$ and $9 = \dfrac{a}{2} + k$; subtract: $4 = a \Rightarrow a = 4$; then $9 = 2 + k \Rightarrow k = 7$